

P.1. Fluidos: Corrientes de gravedad

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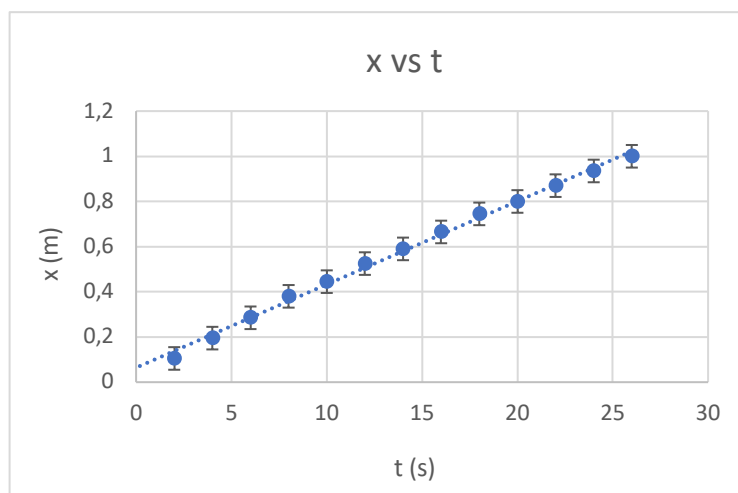
Velocidades de corriente “U”:

Ecuaciones de ajuste tipo ($y = mx + n$).

$d = 6 \text{ cm}$:

$\Delta\rho/\rho_1 = 1\%$:

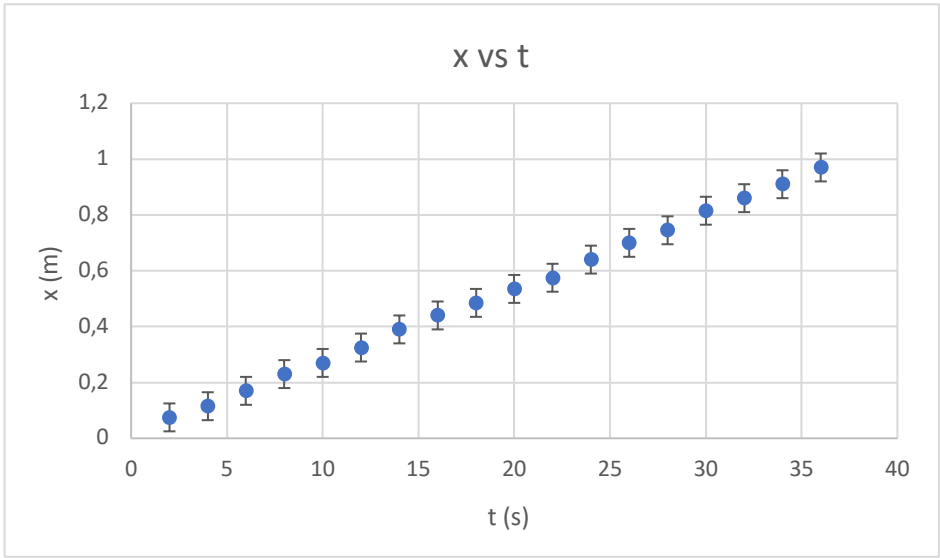
$t (\pm 0.1 \text{ s})$	$x (\pm 0,005 \text{ m})$
2,0	0,105
4,0	0,195
6,0	0,285
8,0	0,380
10,0	0,445
12,0	0,525
14,0	0,590
16,0	0,665
18,0	0,745
20,0	0,800
22,0	0,870
24,0	0,935
26,0	1,000



$m \text{ (m/s)}$	$n \text{ (m)}$	R^2
0.037 ± 0.004	0.06 ± 0.06	0,998

$\Delta\rho/\rho_1 = 0.5\%$:

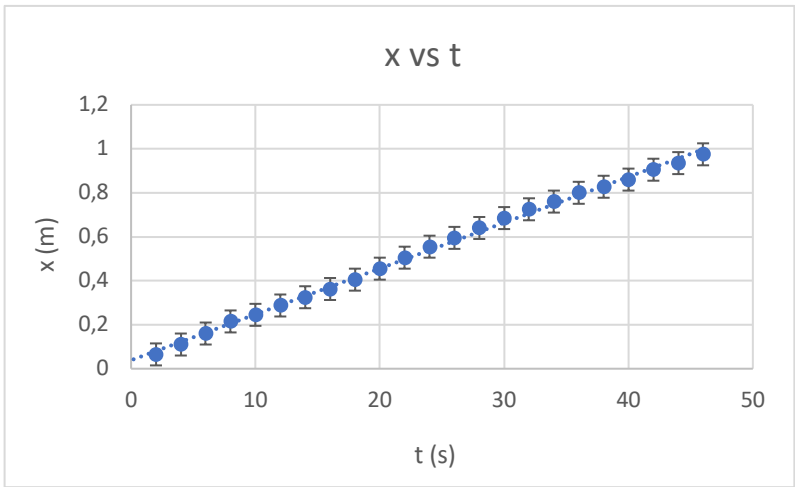
t (± 0.1 s)	x (± 0,005 m)
2,0	0,075
4,0	0,115
6,0	0,170
8,0	0,230
10,0	0,270
12,0	0,325
14,0	0,390
16,0	0,440
18,0	0,485
20,0	0,535
22,0	0,575
24,0	0,640
26,0	0,700
28,0	0,745
30,0	0,815
32,0	0,860
34,0	0,910
36,0	0,970



m (m/s)	n (m)	R ²
0.0260 ± 0.0020	0.01 ± 0.05	0,999

$\Delta\rho/\rho_1 = 0.25\%$:

t (± 0.1 s)	x (± 0,005 m)
2,0	0,065
4,0	0,110
6,0	0,160
8,0	0,215
10,0	0,245
12,0	0,287
14,0	0,325
16,0	0,362
18,0	0,405
20,0	0,455
22,0	0,505
24,0	0,555
26,0	0,595
28,0	0,640
30,0	0,685
32,0	0,725
34,0	0,760
36,0	0,800
38,0	0,827
40,0	0,860
42,0	0,905
44,0	0,935
46,0	0,975

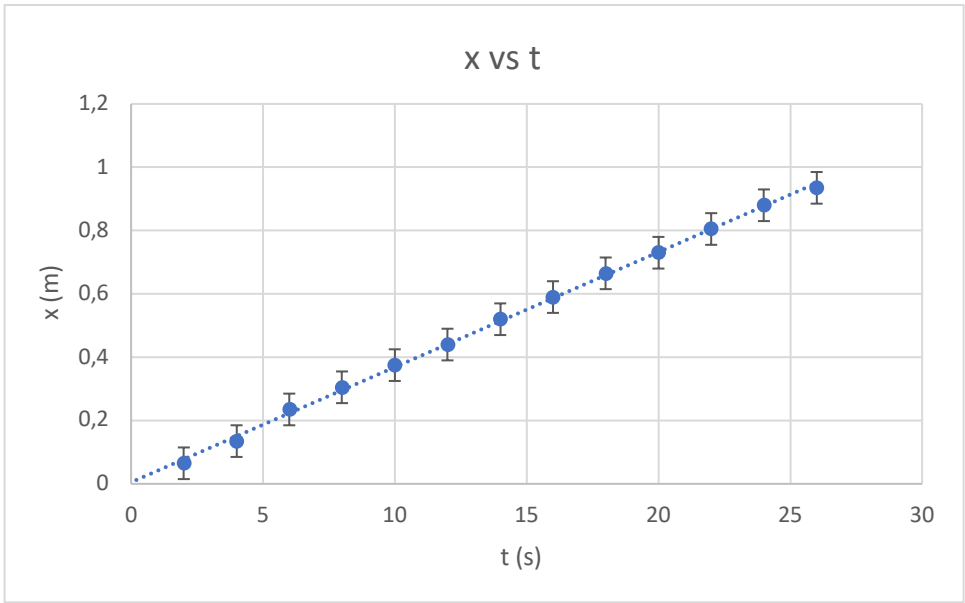


m (m/s)	n (m)	R ²
0.0208 ± 0.0015	0,04 ± 0,04	0,998

d = 8 cm:

$\Delta\rho/\rho_1 = 1\%$:

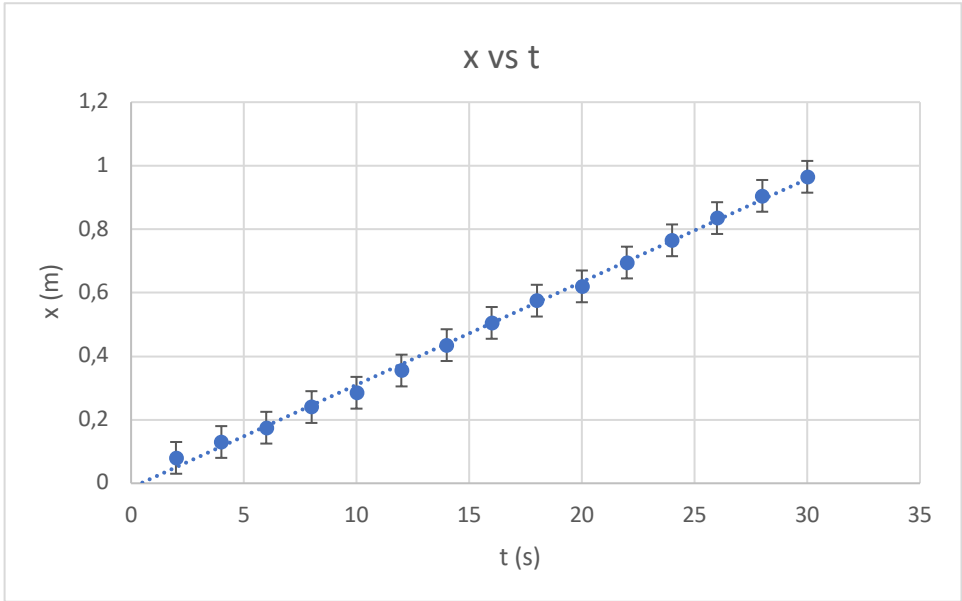
t (± 0.1 s)	x (± 0,005 m)
2,0	0,065
4,0	0,135
6,0	0,235
8,0	0,305
10,0	0,375
12,0	0,440
14,0	0,520
16,0	0,590
18,0	0,665
20,0	0,730
22,0	0,805
24,0	0,880
26,0	0,935



m (m/s)	n (m)	R ²
0.036 ± 0.003	0.00 ± 0.06	0,999

$\Delta\rho/\rho_1 = 0,5\%$:

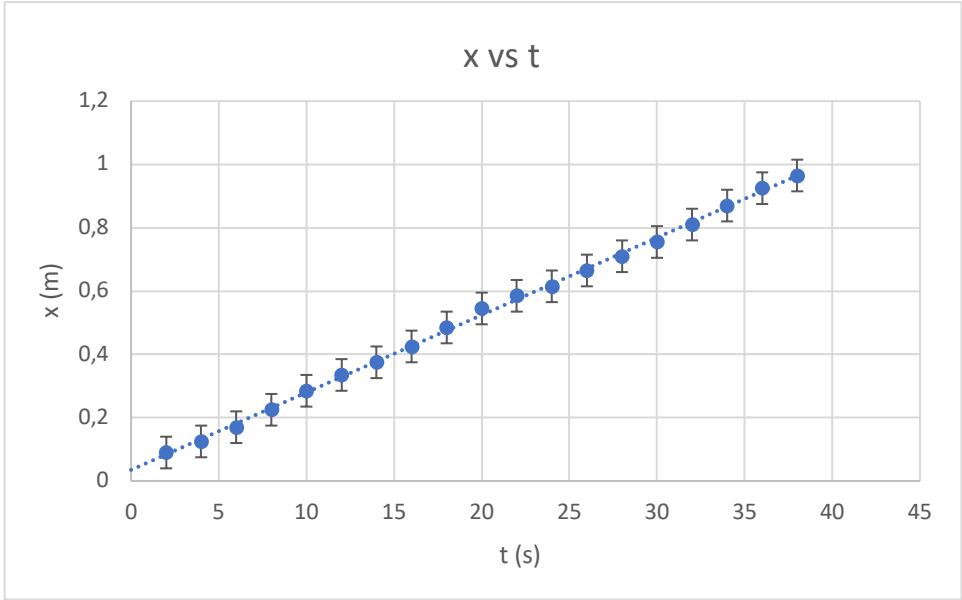
t (± 0.1 s)	x (± 0,005 m)
2,0	0,115
4,0	0,165
6,0	0,210
8,0	0,275
10,0	0,320
12,0	0,390
14,0	0,470
16,0	0,540
18,0	0,610
20,0	0,655
22,0	0,730
24,0	0,800
26,0	0,870
28,0	0,940
30,0	1,000



m (m/s)	n (m)	R ²
0.032 ± 0.003	- 0.01 ± 0.05	0,997

$\Delta\rho/\rho_1 = 0,25\%$:

t (± 0.1 s)	x (± 0,005 m)
2,0	0,090
4,0	0,125
6,0	0,170
8,0	0,225
10,0	0,285
12,0	0,335
14,0	0,375
16,0	0,425
18,0	0,485
20,0	0,545
22,0	0,585
24,0	0,615
26,0	0,665
28,0	0,710
30,0	0,755
32,0	0,810
34,0	0,870
36,0	0,925
38,0	0,965

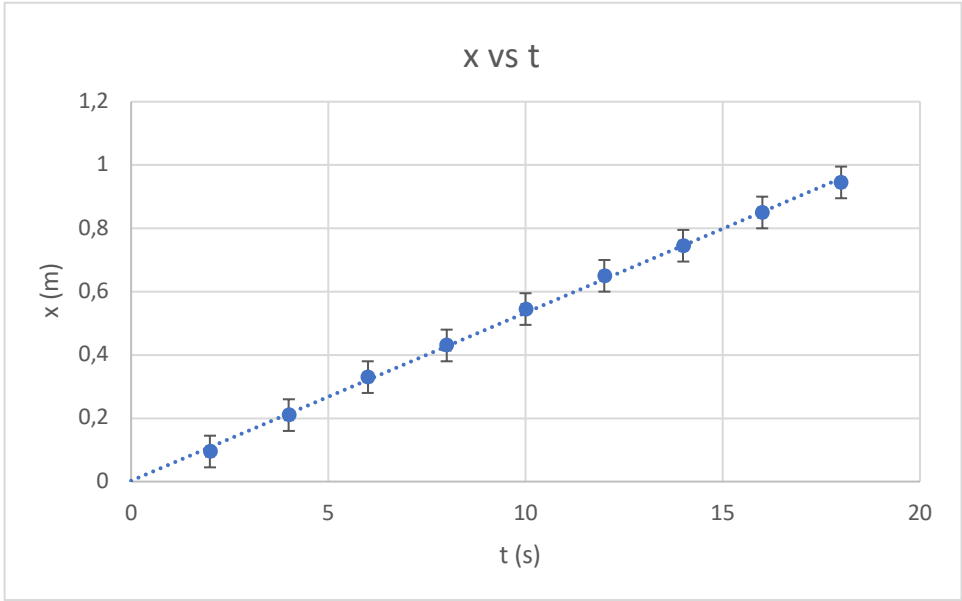


m (m/s)	n (m)	R ²
0.0240 ± 0.0020	0.04 ± 0.05	0,998

d = 12 cm:

$\Delta\rho/\rho_1 = 1\%$:

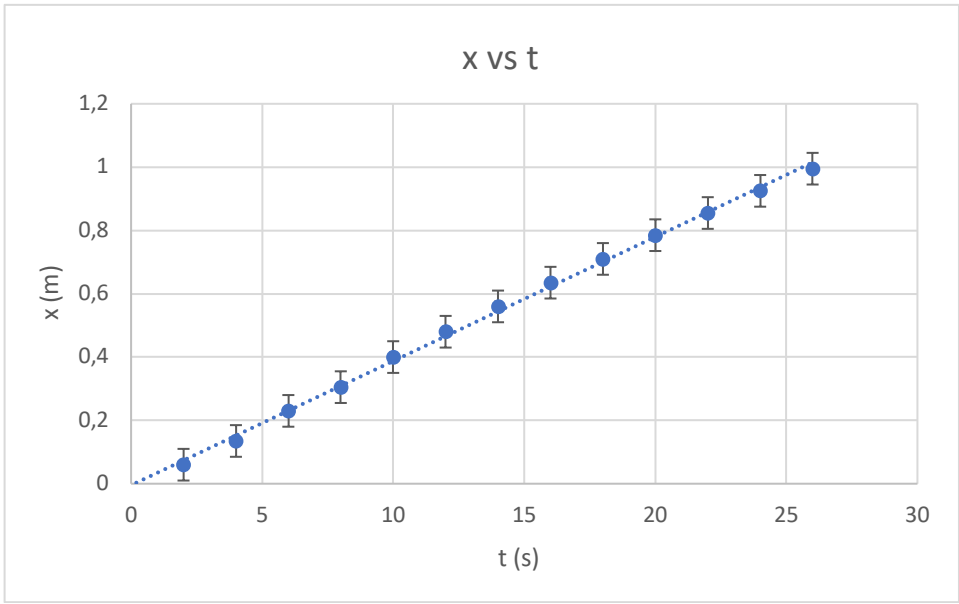
t (± 0.1 s)	x (± 0,005 m)
2,0	0,095
4,0	0,210
6,0	0,330
8,0	0,430
10,0	0,545
12,0	0,650
14,0	0,745
16,0	0,850
18,0	0,945



m (m/s)	n (m)	R ²
0.053 ± 0.006	0.00 ± 0.06	0,999

$\Delta\rho/\rho_1 = 0,5\%$:

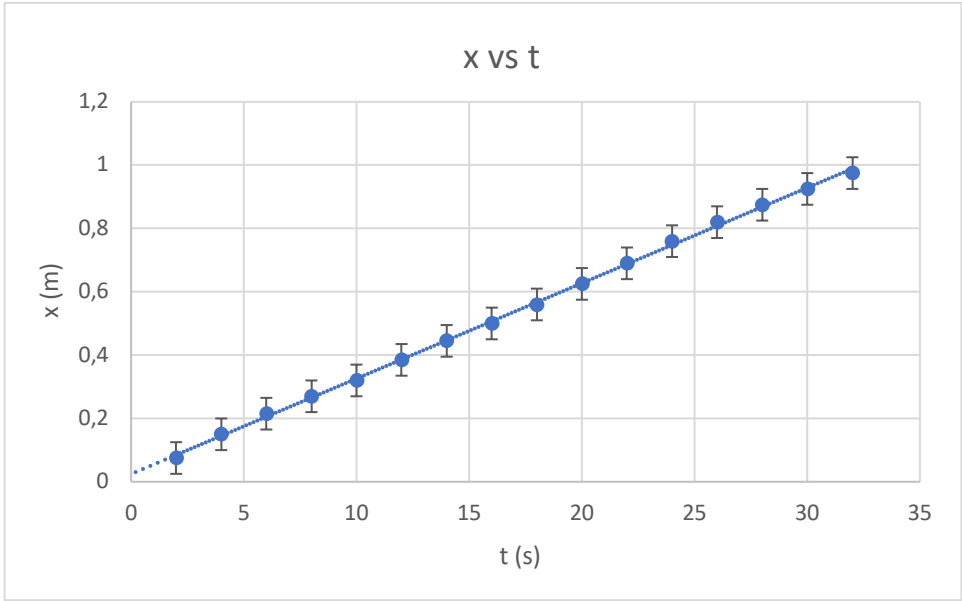
t (± 0.1 s)	x (± 0,005 m)
2,0	0,060
4,0	0,135
6,0	0,230
8,0	0,305
10,0	0,400
12,0	0,480
14,0	0,560
16,0	0,635
18,0	0,710
20,0	0,785
22,0	0,855
24,0	0,925
26,0	0,995



m (m/s)	n (m)	R ²
0.039 ± 0.004	0.00 ± 0.06	0,999

$\Delta\rho/\rho_1 = 0,25\%$:

t (± 0.1 s)	x (± 0,005 m)
2,0	0.075
4,0	0.150
6,0	0.215
8,0	0.270
10,0	0.320
12,0	0.385
14,0	0.445
16,0	0.500
18,0	0.560
20,0	0.625
22,0	0.690
24,0	0.760
26,0	0.820
28,0	0.875
30,0	0.925
32,0	0.975

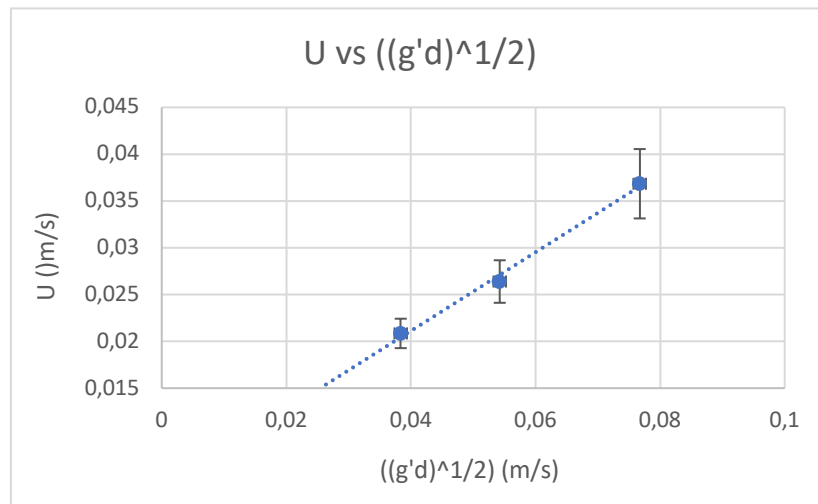


m (m/s)	n (m)	R ²
0.030 ± 0.003	0.02 ± 0.05	0,999

U vs $((g' \cdot d)^{1/2})$:

d = 6 cm:

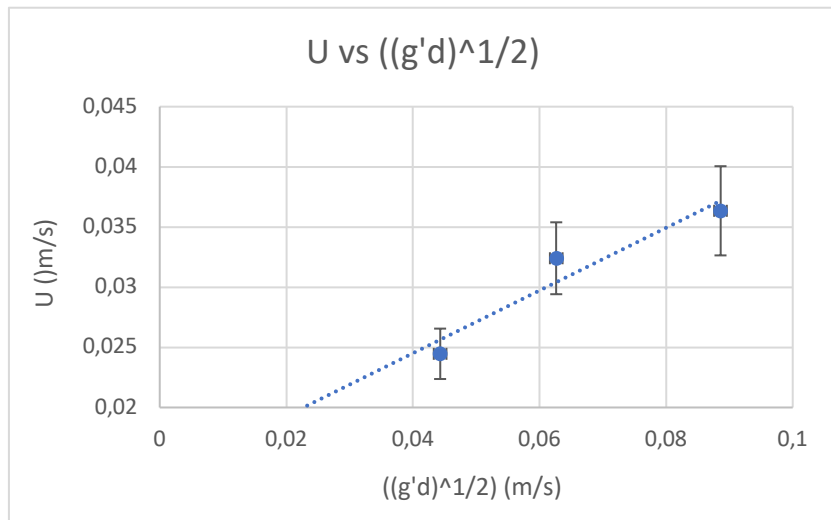
$(\Delta\rho/\rho)$	$g' \text{ (m/s}^2\text{)}$	$((g' \cdot d)^{1/2} \pm 0.001 \text{ m/s})$	U (m/s)
0.01	0.0981	0.077	0.037 ± 0.004
0.005	0.0490	0.054	0.0265 ± 0.0020
0.0025	0.0245	0.038	0.0208 ± 0.0015



m	n (m/s)	R^2
0.40 ± 0.09	0.005 ± 0.004	0.997

d = 8 cm:

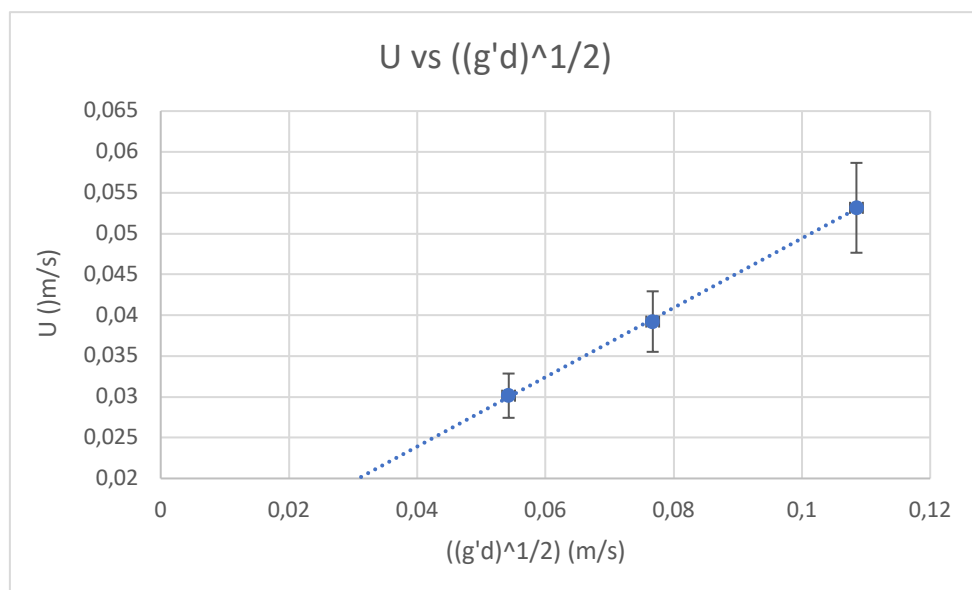
$(\Delta\rho/\rho)$	$g' \text{ (m/s}^2\text{)}$	$((g' \cdot d)^{1/2} \pm 0.001 \text{ m/s})$	U(m/s)
0.01	0.0981	0.088	0.036 ± 0.004
0.005	0.0490	0.063	0.032 ± 0.003
0.0025	0.0245	0.044	0.0245 ± 0.0020



m	n (m/s)	R ²
0,28 ± 0,09	0,012 ± 0,005	0,958

d = 12 cm:

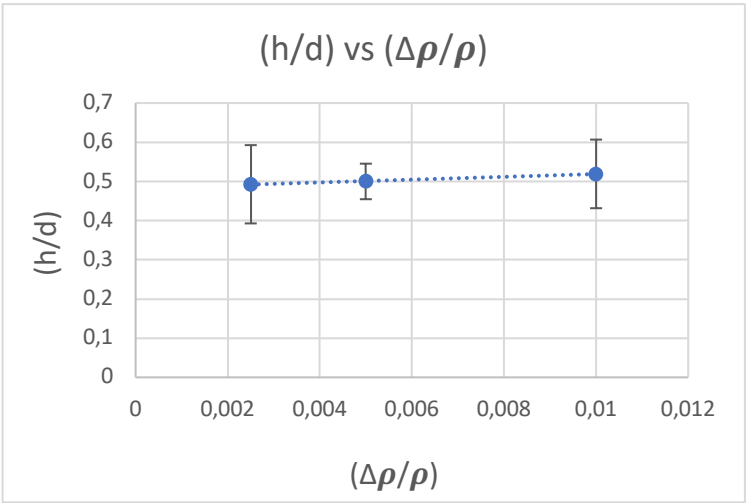
$(\Delta\rho/\rho)$	g' (m/s ²)	$((g' \cdot d)^{1/2} \pm 0.001 \text{ m/s})$	U(m/s)
0.01	0.0981	0.108	0.053 ± 0.006
0.005	0.0490	0.076	0.039 ± 0.004
0.0025	0.0245	0.054	0.030 ± 0.003



m	n (m/s)	R ²
0,42 ± 0,10	0,007 ± 0,007	0,999

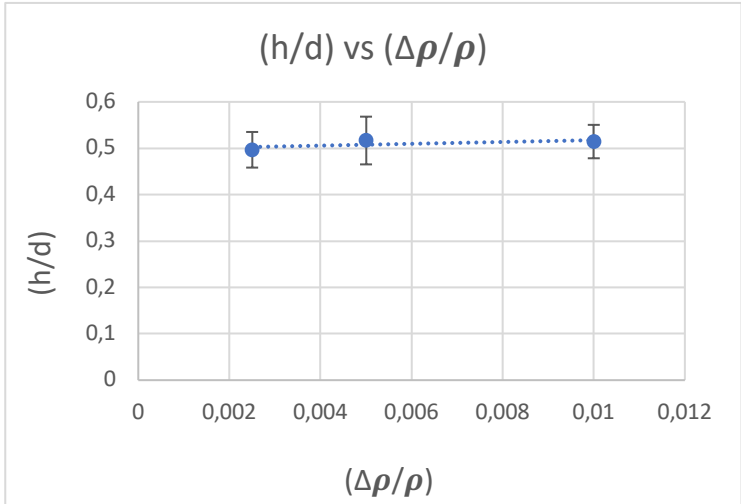
(h/d) vs ($\Delta\rho/\rho$):

d = 6 cm:



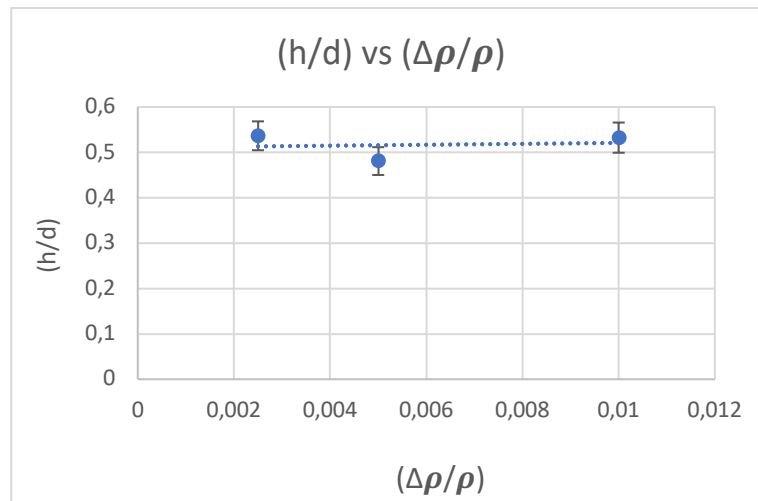
m	n	R ²
4 ± 16	0.50 ± 0.10	0,997

d = 8 cm:



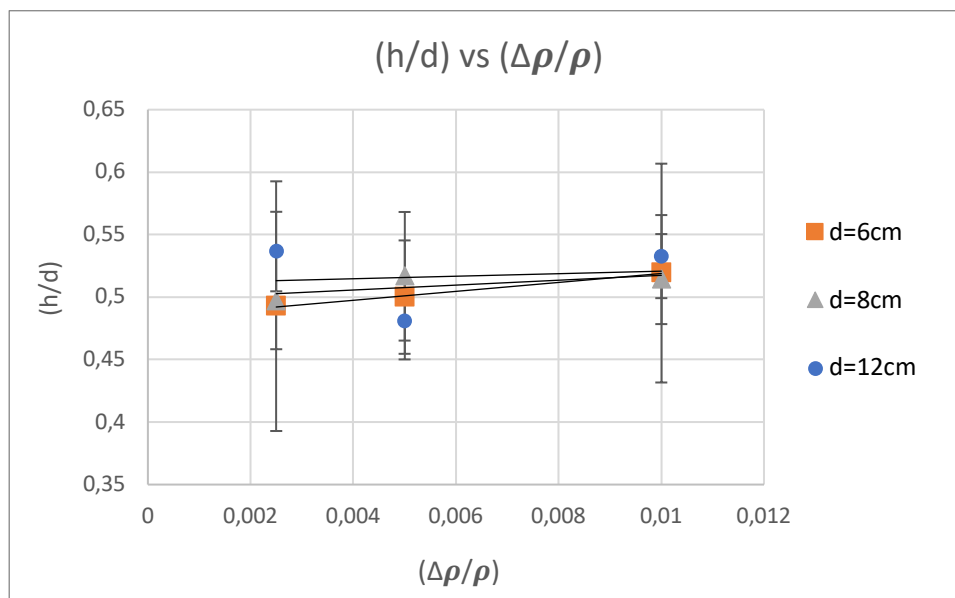
m	n	R ²
2 ± 7	0.50 ± 0.05	0,684

d = 12 cm:



m	n	R ²
1 ± 6	0.50 ± 0.04	0,124

Comparación:



En teoría, la pendiente de las tres rectas debería ser nula puesto que la relación “ (h/d) ” se tiene que mantener constante, y en el valor de “0.5”.